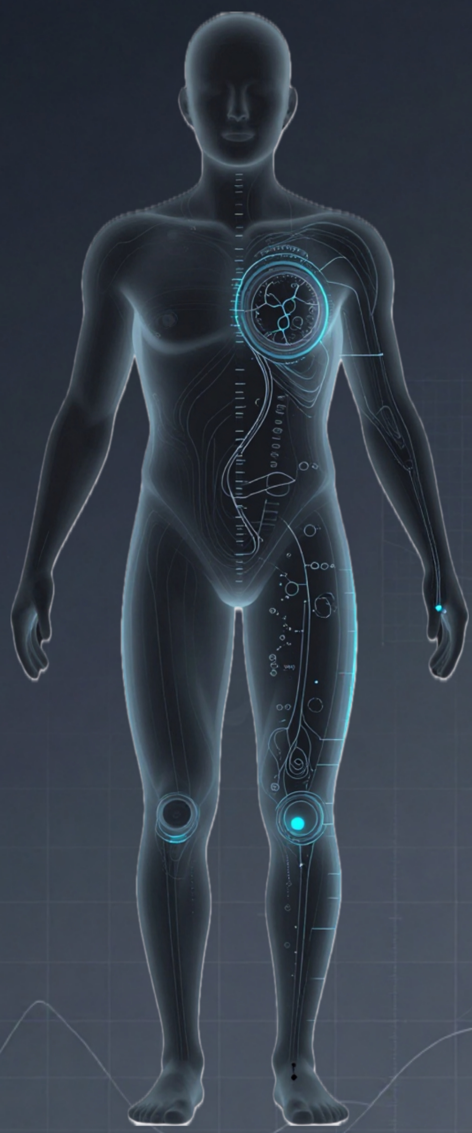
BODY RECOMPOSITION MANUAL



RE/BUILT HEALTH

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CHAPTER 1

PHYSIOLOGICAL AND METABOLIC FOUNDATIONS

Introduction

The fitness and commercial nutrition industry has saturated the market with dogmas, arbitrary restrictions, and promises of rapid results that ignore human biology. The result is a cycle of temporary weight loss followed by metabolic rebound and loss of muscle mass. Body recomposition and health management do not depend on eliminating food groups or following passing trends, but rather on understanding and applying thermodynamics and physiology.

Before calculating macronutrients or structuring a caloric deficit, it is important to understand how the body processes energy at the cellular level. By mastering these principles, you can eliminate anxiety around food and regain control, basing every decision on quantifiable data and clinical evidence.

Physiology of Energy Expenditure and the Body's Actual Energy Requirements

The human body requires a constant flow of energy to maintain homeostasis, repair tissues, and perform mechanical work (movement). This energy is obtained from the chemical bonds in food, which are broken down through cellular respiration to produce Adenosine Triphosphate (ATP) (Guyton & Hall, 2021).

The body's actual requirements do not respond to dietary ideologies, but to biological demands structured around the partitioning of macronutrients. Each macronutrient serves a specific physiological role:

- **Protein (Amino Acids):** Its need is constant for the synthesis of muscle tissue, the production of metabolic enzymes, the transport of molecules in the bloodstream, and the maintenance of the immune system (Schoenfeld, 2020).

- **Lipids (Fats):** They are essential for cellular membrane integrity and act as precursors for crucial steroid hormones such as testosterone and cortisol. They are also the primary energy substrate during rest and low-intensity cardiovascular activity and enable the absorption of fat-soluble vitamins (A, D, E, and K).

- **Carbohydrates:** They are the most efficient energy substrate. They are stored as glycogen in the muscles and liver, providing rapidly available energy for the central nervous system and for high-intensity muscle contractions (resistance training or intervals).

The Insulin and Sugar Myth

One of the most widespread and erroneous dogmas in modern nutrition is the claim that increases in body fat are caused exclusively by insulin spikes, and that this hormone rises only when consuming sugars or simple carbohydrates.

Insulin is a hormone that manages nutrients in the bloodstream and directs them toward cells in the liver, muscles, and adipose tissue.

Clinical evidence demonstrates that carbohydrates are not the only macronutrient that stimulates insulin secretion. Amino acids in the bloodstream directly stimulate the pancreas to release insulin, which is the hormone responsible for “opening” the muscle cell so that amino acids can enter and repair tissue after training (Holt et al., 1997; Power et al., 2009).

Physiological stress, such as intense training or insufficient sleep, increases cortisol and catecholamines, which induce the liver to release glucose into the bloodstream.

Fat storage occurs only when there is a prolonged energy surplus over time, regardless of whether that energy comes from cake, excess olive oil, or excessive protein intake (Hall et al., 2012).

Metabolism: The Chemical Engine of the Body

It is common to hear statements such as “I have a slow metabolism,” “I damaged my metabolism,” or “you need to speed up your metabolism with this supplement.” These concepts treat metabolism as though it were a muscle that can atrophy or a machine that can break down. Metabolism is simply the total sum of all the chemical reactions occurring in your body every second to keep you alive.

Imagine your body as a factory operating 24 hours a day. For this factory to operate, it must process raw materials (food) to generate energy, repair worn machinery (muscles and organs), and eliminate waste. All of this continuous activity is metabolism.

This large-scale process is divided into two phases that occur simultaneously:

- **Catabolism:** When you eat a chicken breast, your body does not use it as-is; catabolism breaks it down into individual amino acids and releases energy in the process. When you are in a caloric deficit, the body uses catabolic processes involving adipose tissue (fat) to use it as fuel.

- **Anabolism:** Anabolism takes those amino acids from the chicken and assembles them to create new muscle tissue in your biceps or legs after training. This process requires a significant amount of energy. This is why building muscle is metabolically “expensive” and requires sufficient calories and protein.

The “Slow Metabolism” Myth and Metabolic Damage

One of the most common excuses for failing to lose weight is blaming a “slow metabolism” on genetics. However, people with greater body weight or a higher percentage of body fat have a higher basal metabolic expenditure, not a lower one (Ravussin et al., 1986). A larger body requires more energy simply to exist, pump blood, and move.

What people often confuse with “metabolic damage” is actually a normal and documented biological process known as adaptive thermogenesis or metabolic adaptation (Trexler et al., 2014).

When you reduce your food intake and lose weight, your body interprets this as a threat to survival (your reptilian brain believes there is a food shortage in the environment). As a defense mechanism, your body becomes more efficient:

1. You are lighter, so you burn fewer calories when moving.

2. Your nervous system subtly reduces involuntary movements (blinking, moving your hands while talking, changing posture), a concept known as NEAT.

3. Your internal organs slightly reduce their energy expenditure.

Understanding that metabolism is dynamic rather than a genetic sentence is the first step toward mastering your body recomposition.

Energy Requirements and Activity

Once we understand that metabolism is simply the collection of chemical processes that keep us alive, we need to quantify how much energy that “factory” requires to operate each day.

This total is known as Total Daily Energy Expenditure (TDEE). To achieve body recomposition, you must understand and manage the four components that mathematically make up your TDEE:

1. Basal Metabolic Rate (BMR)

It represents between 60% and 70% of all the energy you burn in a day. It is the caloric cost of existing. A person with greater body mass, whether muscle or fat, will have a higher BMR simply because there is more biological tissue to keep alive.

2. Non-Exercise Activity Thermogenesis (NEAT)

NEAT represents the energy expended on everything that is not sleeping, eating, or structured exercise, such as walking to work or shopping. It represents approximately 15% to 20% of your daily expenditure. When you enter a caloric deficit, the body subconsciously reduces NEAT (Levine, 2002).

3. Thermic Effect of Food (TEF)

Your body expends energy while chewing, digesting, absorbing, and storing the food you consume. It represents approximately 10% of TDEE, and not all macronutrients require the same amount of energy to digest.

* Fats: low TEF (0–3%).
* Carbohydrates: moderate TEF (5–10%).
* Protein: extremely high TEF (20–30%).

This means that if you consume 100 calories from chicken breast or whey isolate, your body will spend up to 30 of those calories simply processing them (Westerterp, 2004). A high-protein diet not only protects muscle mass, but also mathematically increases your daily energy expenditure without requiring any additional movement.

4. Exercise Activity Thermogenesis (EAT)

This is the caloric expenditure derived from exercise. For most people, it represents only approximately 5% to 10% of total daily energy expenditure.

CHAPTER 2

METABOLIC QUANTIFICATION AND MACRONUTRIENT ANALYSIS

Introduction

To force the body to recompose its structure—to simultaneously oxidize adipose tissue while synthesizing or retaining muscle tissue—it is necessary to move away from intuitive estimation of food intake.

We will address how to calculate your energy variables algorithmically (BMR and TDEE) and the expected protein distribution according to activity level.

What Macronutrients Are and How the Body Uses Them

Traditionally, macronutrients are divided into three main categories, whose intrinsic energy values are calculated using the classic Atwater factors (Merrill & Watt, 1973): protein (4 kcal/g), carbohydrates (4 kcal/g), and lipids or fats (9 kcal/g). Alcohol, although not an essential nutrient, provides 7 kcal/g, and the body prioritizes it as a toxin that must be oxidized before other substrates.

1. Protein: The Structural Substrate

The body does not use protein as its primary fuel source unless it is in a state of severe starvation. Once digested, proteins are broken down into amino acids that enter the bloodstream.

The body maintains a circulating pool of amino acids that is used primarily for building and repair processes: muscle protein synthesis (MPS), tissue regeneration, and the production of enzymes and peptide hormones. The purpose of maintaining a high protein intake, particularly during a caloric deficit, is to promote a positive nitrogen balance. This provides a powerful biological signal that helps prevent muscle tissue catabolism, directing the body toward its fat stores as an energy source (Antonio et al., 2015).

2. Carbohydrates: The High-Performance Substrate

Their primary and almost exclusive physiological function is to provide rapidly available energy. After digestion, carbohydrates are converted into glucose. Some circulates in the bloodstream to supply the central nervous system—the brain is a major consumer of glucose—while the excess is stored as glycogen.

Glycogen has two primary storage sites: the liver, which uses it to maintain stable blood glucose levels between meals, and skeletal muscle, which uses it as an energy source during high-intensity muscular contractions (Murray & Rosenbloom, 2018). Carbohydrates are essentially the fuel that allows you to sustain resistance training or high-density workouts and maintain the anabolic stimulus in your muscles.

3. Fats (Lipids): The Regulatory and Storage Substrate

Their critical role is both endocrine and cellular. Dietary fats—particularly cholesterol and fatty acids—are fundamental components of cell membranes and act as chemical precursors for the synthesis of important steroid hormones such as testosterone, estrogen, and cortisol. A drastic reduction in dietary fat below a minimum biological threshold can significantly impair endogenous hormone production, interfere with the absorption of fat-soluble vitamins, and hinder metabolic progress (Venkatraman et al., 2000).

Why Adequate Protein Intake Matters

The Recommended Dietary Allowance (RDA) for protein is approximately 0.8 grams per kilogram of body weight. This figure represents the minimum amount necessary to prevent nutritional deficiencies and muscle tissue loss in sedentary individuals (Phillips et al., 2016).

When the goal is body recomposition, protein becomes the primary tool within the system. Its importance is based on three pillars:

1. **Preservation of Lean Tissue — Anabolic Signaling:** During a caloric deficit, if protein intake is not high enough to maintain a constant supply of amino acids in the bloodstream, the body will break down muscle tissue to obtain those amino acids.

2. **Thermodynamic Advantage — Thermic Effect:** Protein digestion requires the body to expend approximately 20–30% of the calories it provides.

3. **Hunger Control — Endocrine Satiety:** Protein is the most satiating macronutrient. Its consumption delays

gastric emptying and stimulates the release of appetite-suppressing hormones such as GLP-1 and peptide YY, while reducing ghrelin, the hunger hormone (Paddon-Jones et al., 2008).

The Algorithmic Calculation: How Much Protein to Consume

To optimize body recomposition, protein intake should be calculated according to the individual's body mass. Mathematical thresholds are established according to the person's energy state (Morton et al., 2018).

Standard Protocol — Maintenance or Mild Surplus: For individuals seeking hypertrophy or looking to maintain their weight while optimizing body composition, the optimal range is:

$$\textbf{Daily Protein} = \textbf{Total Body Weight (kg)} \times \textbf{1.6–2.2 g}$$

Consuming more than 2.2 g/kg in this state does not provide significant additional benefits for muscle protein synthesis, although it is not harmful.

Caloric Deficit Protocol — Fat Loss

As the caloric deficit becomes more aggressive, the risk of losing muscle mass increases substantially. The more precise calculation is based on Lean Body Mass (LBM) (Helms et al., 2014).

$$\textbf{Daily Protein} = \textbf{LBM (kg)} \times \textbf{2.3–3.1 g}$$

For example, an individual weighing 80 kg with 20% body fat has a Lean Body Mass of 64 kg. If this individual enters a caloric deficit to reduce body fat to 12%, their daily protein requirement would be:

$$\textbf{64 kg LBM} \times \textbf{2.5 g} = \textbf{160 g of protein per day}$$

Energy Quantification: Basal Metabolic Rate (BMR) and Dynamic Total Daily Energy Expenditure (TDEE)

To design a caloric deficit that eliminates fat without destroying muscle mass, it is necessary to establish your actual energy starting point.

In the Re/Built Health system, we use a dynamic quantification model that adds the physiological cost of maintaining your body at rest to the energy expenditure generated by your daily activity.

Basal Metabolic Rate (BMR)

Basal Metabolic Rate, or BMR, is the amount of energy, measured in kilocalories, that your body requires to maintain its vital functions at complete rest, including respiration, circulation, and neurological function.

To calculate it, we use the Katch-McArdle equation, which is based exclusively on Lean Body Mass (LBM) (McArdle et al., 2015).

Universal Formula

BMR = 370 + (21.6 × LBM in kg)

This is the amount of energy you would burn if you spent 24 hours in bed without moving a muscle.

Total Daily Energy Expenditure (TDEE)

Traditional nutrition calculates Total Daily Energy Expenditure (TDEE) by multiplying BMR by a fixed “activity factor”—for example, multiplying it by 1.55 if you perform moderate exercise. But life is not linear. One day you may perform a hard workout and drive home afterward. The next day you may not train at all, but you may have to walk quickly because you are running late or use public transportation. Your actual TDEE for each day is composed of:

Dynamic TDEE = BMR + Active Calories

The Real-Effort Algorithm

Devices such as the Apple Watch use a system that combines a three-axis accelerometer and gyroscope with photoplethysmography—optical sensors that measure heart rate (Shcherbina et al., 2017).

The algorithm is designed to filter out “noise.” If you constantly move your hand while sitting at your desk, the watch detects the movement, but because there is no corresponding increase in cardiovascular demand or actual displacement of your body through space, it does not register it as significant physical effort and therefore does not add significant active calories.

For the telemetry to be operationally useful, the technology must be used consistently:

NEAT Validation: When you walk quickly to work, the watch detects continuous displacement and a slight increase in heart rate, adding those active calories with a high degree of precision (Düking et al., 2020).

Strict Use of Workout Mode: For structured exercise data to be reliable, you must indicate the type of activity to your device. If you are performing a functional circuit, weight training, or jogging, activate the corresponding workout mode. This calibrates the sensors to correctly interpret the cardiovascular demand and mechanical work associated with the specific training session (Achten & Jeukendrup, 2003).

Isolating Effort: Understand that active calories represent the energy cost of overcoming resistance or moving your body.

At the end of each day, your device's health application will provide you with a total number of Active Calories. Add these to your BMR to obtain your TDEE.

CHAPTER 3

INTAKE ORGANIZATION AND MEASUREMENT METHODOLOGY

Introduction

The human brain is poor at estimating volumes and portions visually (Chandon & Wansink, 2007). To achieve predictable, mathematical results, eating must stop being an intuitive act and become a data-auditing process.

Tools Required to Organize Your Intake

To execute this system, you need to standardize your tools. Trying to achieve body recomposition without accurate measurement tools is equivalent to trying to build a building without a measuring tape.

1. Digital Kitchen Scale

Volume measurement is one of the greatest sources of caloric error. A “cup of oats” can range from 70 to 110 grams depending on how compacted it is. A “tablespoon of peanut butter” can contain 15 or 35 grams, representing a difference of more than 100 kilocalories in a single serving. The only valid metric standard is mass, measured in grams.

Technical requirement: A digital kitchen scale with 1-gram precision and a tare function. This will be your most frequently used tool.

2. Food Storage Containers

If you have to decide what to cook, measure it, and prepare it three times a day, your cognitive capacity will eventually become exhausted and you will end up breaking the system. Use airtight glass or BPA-free plastic containers of a uniform size. Batch organization (cooking large quantities of carbohydrates and protein separately for three or four days) allows you to weigh food once in large batches and then divide the portions mathematically.

How to Organize What You Eat: There Are No Restrictions, It Depends on Your Energy Expenditure

The diet industry profits from selling lists of “allowed” and “forbidden” foods. Carbohydrates are demonized, dairy products are prohibited, and eating times are restricted. From the perspective of thermodynamics and human biology, food has no moral value. Foods are not “good” or “bad,” nor do they have the ability to make you store fat simply because you consume them.

Food is simply a vehicle for delivering macronutrients (protein, carbohydrates, and fats) and micronutrients (vitamins and minerals).

Your Total Daily Energy Expenditure (dynamic TDEE) establishes your daily limit. As long as your calories fit within that limit and you maintain the required protein intake to preserve lean mass, you can structure your meals with complete freedom. This approach is known as Flexible Dieting, or IIFYM (If It Fits Your Macros).

Rigid dietary approaches (eliminating entire food groups) generate high levels of psychological stress, encourage binge eating, and have statistically higher long-term failure and rebound rates (Stewart et al., 2002). By contrast, flexible dieting improves adherence and the relationship with food without compromising fat loss (Conlin et al., 2021).

You are not restricted to eating boiled chicken and broccoli seven times a day. If your active energy expenditure was particularly high today because you trained legs and took long walks, your dynamic TDEE will be higher. Consequently, your carbohydrate "budget" for that day increases, allowing you to consume larger amounts of food. Body recomposition is about managing data, not suffering.

Calculate Your Macros and Calories (Using Free AI Tools or Calculation Apps)

Knowing how much protein you need and what your TDEE is serves no purpose if you do not quantify what you actually put on your plate. You no longer need printed nutritional-value tables or to perform calculations by hand. The process should be fast, digital, and frictionless.

1. Using Mobile Databases (Tracking Apps)

The most standard and efficient method is to use mobile nutrition-tracking applications. The most accurate apps, with the largest databases, include MacrosFirst, Cronometer, FatSecret, and MyFitnessPal.

2. Using Artificial Intelligence (ChatGPT, Claude, Gemini)

When you cook complex recipes, bake high-protein products, or combine multiple ingredients—such as a protein loaf made in an air fryer—adding each ingredient individually to an app can be tedious. You can provide an AI system with the exact list of raw ingredients and ask it to provide the total nutritional breakdown or the breakdown per serving.

Technical AI Prompt Example

"I have prepared a recipe using the following raw ingredients: 150 g of oat flour, 60 g of whey isolate protein, 2 large whole eggs, 100 ml of skim milk, and 10 g of pure cocoa powder. The entire mixture weighs 400 g after cooking. Divide it into 4 equal portions. Give me the exact total calories and macronutrients (protein, carbohydrates, and fats) for each 100 g serving."

The AI will calculate the standard nutritional values and return the corresponding nutritional profile. You can even create an assistant that records your daily intake and provides you with the total at the end of each day.

By automating this process with technology, eating loses much of its emotional burden and becomes a matter of applied mathematics. Once you develop the habit of weighing your food and entering the data (which takes less than five minutes per day in total) you gain complete control over your body composition.

CHAPTER 4

OPERATION OF THE QUANTIFICATION SYSTEM (CALCULATION TOOL)

Introduction

Re/Built Health operates through a structured spreadsheet designed for a 365-day record that calculates and projects your physiological variables. It is essential to understand how and where to enter the data correctly.

Measurements: Basic Anthropometry and Data Collection

Total body weight fluctuates daily due to water retention, muscle glycogen stores, sodium intake, stress (cortisol), and intestinal transit. For the purposes of Re/Built Health, body weight is only a raw variable. For the system to determine your Lean Body Mass (LBM) and apply the Katch-McArdle thermodynamic equation, it needs to isolate your lean mass from adipose tissue.

Because undergoing weekly clinical DEXA scans is technically and financially impractical, the tool uses validated anthropometric algorithms. The system processes the U.S. Navy Method and the Relative Fat Mass (RFM) Index, which have demonstrated a high correlation with laboratory methods for estimating body fat in general populations (Hodgdon & Beckett, 1984; Woolcott & Bergman, 2018).

In the Measurements section, you must record the following variables:

1. **Total Body Weight:** In kilograms, including decimals (e.g., 80.5 kg).
2. **Height:** In centimeters.
3. **Neck Circumference:** Measured immediately below the larynx (below the Adam's apple in men). The tape should be parallel to the floor and snug against the skin without compressing it.
4. **Waist Circumference:** Measured at the narrowest point of the abdomen, generally at or just above the navel, while keeping the abdomen relaxed. Do not pull your stomach in or force exhalation.
5. **Hip Circumference (Female Anatomy Only):** Measured at the widest transverse point of the buttocks. Note: The predictive equation for male physiology does not require this measurement because of differences in the genetic distribution of body fat.

Clinical Data Collection Protocol (Standardization)

Measurements must be recorded as follows:

- **Time:** Always in the morning, immediately after waking and after using the bathroom.
- **State:** Completely fasted, before consuming water, coffee, or food.
- **Tool:** Use a flexible body measuring tape.
- **Frequency:** If you want more precise monitoring, you can enter all measurements daily. One measurement per week is sufficient for tracking progress and avoiding unnecessary psychological stress.

Intake: Recording Energy Flow

The Intake section is your daily nutritional record. At the end of each day, you must transfer the final totals provided by your application or AI into the corresponding cells of the tool:

1. **Total Protein (g)**
2. **Total Fat (g)**
3. **Total Carbohydrates (g)**
4. **Total Calories**

The Importance of Accurate Tracking

Consistent self-monitoring is the strongest behavioral predictor of success in modifying body composition (Burke et al., 2011). By requiring you to record your final numbers every night in the calculation matrix, you create a moment of objective accountability.

If your macros deviate from your plan on a given day (because of a social event or a failure in meal preparation), you must enter the actual data anyway. The system does not judge; it is a mathematical algorithm. Omitting a high-intake day corrupts the entire week's data and compromises the tool's ability to function properly.

TDEE: The Dynamic Balance Calculation

The TDEE (Total Daily Energy Expenditure) section of the tool is where physiology meets telemetry technology. This is where the system determines how much energy you actually burned over the previous 24 hours and, by cross-referencing this data with your Intake, reveals your metabolic state (deficit, maintenance, or surplus).

This section operates semi-automatically, combining fixed calculations with the input of your daily data.

1. Variables Automated by the System

Using the anthropometric data you entered in the Measurements section, the tool calculates your Lean Body Mass (LBM). Using this data, the system automatically processes the Katch-McArdle equation to provide your daily Basal Metabolic Rate (BMR). This cell is locked and automated; you do not need to manually calculate your BMR every time your weight changes. The matrix does it for you based on your current body composition.

2. Operational Variable: Active Calorie Input

At the end of your day, open the health application on your device or smartwatch (e.g., Apple Fitness) and locate the Active Calories (or Move Calories) metric. This number represents the energy expenditure derived from your NEAT (walking and daily activity) and your EAT (your sessions recorded using workout mode, such as treadmill intervals or functional circuits).

Enter the number of Active Calories in the designated cell of the tool. The system will execute the following operational equation:

Dynamic Daily TDEE = BMR + Active Calories

Once your Dynamic TDEE has been calculated, the tool will automatically compare it with the Total Calories recorded under Intake, using the energy-balance equation (Hall et al., 2012):

$$\textbf{Balance} = \textbf{Dynamic Daily TDEE} - \textbf{Total Intake}$$

If the number is negative, you are in a quantified deficit and oxidizing adipose tissue. If the number is positive, you are in a surplus. By visualizing this data day after day and week after week, your progress stops being a mystery.

Meaning of the Percentages: Clinical Interpretation of the Data

To operate the tool successfully and avoid sabotaging your own progress through misinterpretation, you must understand exactly what the three main percentages reported by the system mean clinically.

1. Body Fat Percentage (% BF)

This is the indicator automatically calculated by the tool by cross-referencing your circumference measurements (Navy Method/RFM Equation).

Body fat percentage represents the fraction of your total body weight that consists of adipose tissue. If you weigh 80 kg and the tool indicates 20% body fat, this means you have 16 kg of fat mass and 64 kg of Lean Body Mass (LBM).

- **The weight trap:** If your weight drops to 78 kg in one month, but your body fat percentage rises to 22%, it means you lost muscle and retained fat (a catabolic state—the worst possible scenario). If your weight remains static at 80 kg, but your body fat percentage drops to 16%, you have achieved pure body recomposition: you oxidized fat and built muscle simultaneously in the same proportion.

- **Clinical parameters:** For men, a percentage between 10% and 15% represents an athletic and metabolically optimal state. For women, due to the fat that is biologically essential for fertility and endocrine function, the optimal athletic range is 18% to 24% (Gallagher et al., 2000). Falling below essential fat levels (less than 5% in men or 12% in women) suppresses the immune system and disrupts hormone production.

2. Macronutrient Distribution Percentage

Traditional nutrition often prescribes diets based on fixed percentages, which is a serious mathematical error. If you consume 1,500 kcal, 30% from protein equals 112 grams. If you consume 3,000 kcal, 30% equals 225 grams. The percentage is the same, but the biological impact is radically different.

3. Caloric Deficit Percentage (Deficit Magnitude)

When the tool cross-references your Total Energy Intake with your Dynamic TDEE, it will calculate the size of your caloric deficit as a percentage. This number determines the aggressiveness of your metabolic intervention.

- **Conservative Deficit (10%–15%):** This is the ideal range for sustained body recomposition. It allows you to lose adipose tissue gradually without compromising training energy or putting muscle mass at risk.
- **Moderate Deficit (20%–25%):** The clinical standard for accelerated fat loss. It requires strict control of protein intake (approaching 2.8–3.1 g per kg of LBM) to preserve muscle mass (Garthe et al., 2011).

- **Aggressive Deficit (>30%):** Unless applied under strict medical supervision in individuals with class II or III obesity, a deficit of this magnitude will trigger severe adaptive thermogenesis. The body will begin to catabolize muscle tissue for energy, your NEAT will collapse due to lethargy, and progress will stall.

If the tool indicates that you have maintained a deficit greater than 30% for multiple days, you are not being "disciplined"—you are inducing structural damage to your metabolism, and you should immediately increase your intake.

CHAPTER 5

EXECUTING THE RECOMPOSITION

Introduction

Body recomposition is not an acute event or a “21-day challenge”; it is the systematic management of thermodynamic variables over time. Here, we will structure how to take the first step without falling into analysis paralysis, how to apply a caloric deficit in real life, and how to manage the most important and underestimated variable of all: adherence.

How to Start the Recomposition: The Starting Protocol

The most destructive mistake when beginning a physical transformation process is attempting to change every aspect of your life in a single day. Emptying the pantry, starting exhausting training routines, and drastically cutting calories simultaneously creates physiological and psychological stress that virtually guarantees early abandonment. The brain perceives these abrupt changes as a threat to homeostasis (Lally et al., 2010).

Phase 1: Establishing the Baseline (Days 1–3)

Before attempting to change your body, you must know exactly where you are today, both geometrically and metabolically, without value judgments or emotional baggage.

1. **Initial measurement:** On the first morning, while fasting and after using the bathroom, use the measuring tape to record the circumferences of your neck, waist (and hips, in the case of female anatomy). Step onto the scale and record your total body weight.

2. **Data entry:** Transfer these numbers to the Measurements section of your spreadsheet. The tool will automatically determine your Lean Body Mass (LBM) and establish your initial Basal Metabolic Rate (BMR). This is your clinical starting point.

Phase 2: Blind Nutritional Audit (Days 4–10)

During the first week, you will not apply any caloric deficit or change the way you eat. This stage has a single purpose: mastering the instrumentation and discovering your subconscious caloric intake.

1. **Instrumentation:** Use your kitchen scale to weigh everything you eat and drink.

2. **Tracking:** Enter your food into your tracking app (MacrosFirst, MyFitnessPal, or via AI) and, at the end of the day, transfer your total protein, fat, carbohydrate, and calorie intake to the calculation matrix.

3. **Effort tracking:** Make sure to record your daily Active Calories, as reported by your smartwatch, in the TDEE section.

Clinical evidence shows that people tend to underestimate their daily intake by as much as 40% (Lichtman et al., 1992).

By auditing your food without altering it for one week, you will cross-reference your actual Total Intake with your actual Dynamic TDEE. The tool will reveal whether you have been living in a chronic caloric surplus. This confrontation with the numerical reality is the program's true psychological turning point.

Phase 3: Mathematical Intervention and Protein Anchoring (Day 11 Onward)

Once you have mastered the habit of weighing and entering data without it becoming an excessive mental burden, it is time to initiate the recomposition.

1. **Protein anchoring:** Look at your Lean Body Mass (LBM) in the spreadsheet and multiply that number by a factor between 2.3 and 2.5 (as discussed in Chapter 2). This number, expressed in grams, is your new daily target. Before planning any other meal, you must ensure that your daily protein sources meet this requirement.

2. **Applying the caloric deficit:** With your BMR already calculated by the system and your Active Calories being recorded daily, aim for a conservative deficit of 15% to 20% of your total Dynamic TDEE.

3. **Allocating the remaining calories:** Once your protein target has been deducted, distribute the remaining calories between fats (no less than 0.8 g per kg of body weight to safeguard hormone production) and carbohydrates (as fuel for your training).

From this day forward, execution consists of repeating the process. Do not look for changes in the mirror during the first few weeks; the human body requires time to oxidize lipids and synthesize new tissue. Focus exclusively on executing the data and maintaining a negative energy balance in the tool.

Using Caloric Deficit and Protein: The Synergy of Recomposition

The clinical success of body recomposition depends on the simultaneous manipulation of two metabolic levers: energy balance (the deficit) and nutrient partitioning (protein). In applied thermodynamics, a fundamental rule governs the outcomes: the caloric deficit determines how much weight you lose; protein intake determines what type of weight you lose (Leidy et al., 2015).

Applying a caloric deficit without adequate protein anchoring is the mistake that leads to the “skinny fat” phenotype. When energy is insufficient, the body catabolizes skeletal muscle and reduces basal energy expenditure (BMR), inevitably leading to a plateau followed by rebound weight gain.

To execute this synergy algorithmically, three operational guidelines should be followed:

1. Energy Partitioning

When you are in a sustained caloric deficit, your body is in a chronic catabolic state. The objective is to direct that catabolism exclusively toward adipose tissue. By consuming between 2.3 and 3.1 grams of protein per kilogram of Lean Body Mass (LBM), you saturate the bloodstream with amino acids. This keeps the mTOR pathway (responsible for muscle protein synthesis) activated, sending a biological signal that "shields" muscle tissue. Because the body cannot break down muscle tissue due to this chemical barrier, it is forced to oxidize triglycerides stored in adipose tissue to cover the energy deficit (Hector & Phillips, 2018).

2. Modulating the Deficit According to Dynamic TDEE

Because your Total Daily Energy Expenditure (TDEE) is dynamic and fluctuates according to your recorded Active Calories, your deficit should not be a static and restrictive number.

- **High-performance days:** If you have an intense training session (e.g., long-distance running or leg hypertrophy), your Dynamic TDEE will increase significantly. On these days, your total intake should increase to keep the deficit within the conservative 15%–20% range. Do not try to "save" those burned calories; your body needs substrate (carbohydrates) to replenish glycogen and protein to repair tissue.

- **Rest days:** If your watch indicates zero EAT and low NEAT, your TDEE will be lower, approaching your Basal Metabolic Rate. On these days, your total intake automatically decreases to maintain a negative energy balance, but protein intake remains unchanged.

3. Plateau Adjustments

The human body is designed to adapt. If, after three weeks of perfect tracking in your calculation matrix, your average body fat percentage does not decrease, a metabolic adaptation has occurred. The clinical response is not to eliminate carbohydrates, but to adjust the mathematics:

- **Option A:** Increase energy expenditure by deliberately increasing NEAT (for example, adding 2,000 fixed daily steps to your routine).

- **Option B:** Increase the caloric deficit by an additional 5%, reducing calories exclusively from fats and carbohydrates, never from protein.

Adherence, Patience, and Control, Not Restriction

The limiting factor in any metabolic intervention is not physiology, but psychology. Even the most perfectly designed body recomposition strategies fail in practice due to decision fatigue, operational friction, and unrealistic expectations of linear progress.

Mitigating Operational Friction

The Re/Built Health system is designed to externalize the cognitive burden. Relying on willpower or daily motivation is a strategy destined to fail. When the brain must constantly decide what to eat, how much to eat, and whether something is “allowed,” neurological exhaustion occurs (decision fatigue).

By centralizing calculations in an automated matrix (the 365-day spreadsheet), eating ceases to be a moral dilemma and becomes a simple matter of data execution. You do not decide whether you are progressing; the numbers do. This systematization dramatically reduces friction, allowing long-term adherence to be maintained in an almost mechanized manner (Wing & Phelan, 2005).

The Fallacy of Absolute Restriction

Severe restriction generates compensatory endocrine responses. Prohibiting specific foods increases cortisol (the stress hormone) and alters dopaminergic reward signaling in the brain, leading to episodes of binge eating.

Real control is not avoiding a slice of pizza out of fear; absolute control is eating that slice of pizza, accurately auditing it, recording it in the calculation tool, deducting it from your Dynamic TDEE, and understanding that the physiological recomposition process remains intact because energy balance and protein anchoring were maintained. This is clinical management, not punitive dieting.

Patience and Nonlinear Fluctuation

Cellular tissue takes time to rebuild. Demanding visible results every week means disregarding biology. Body weight fluctuates daily and unpredictably due to:

- **Glycogen stores:** Each gram of carbohydrate stored in muscle retains between 3 and 4 grams of water.

- **Muscle inflammation:** Resistance training generates microtears that cause localized fluid retention during the repair process.

- **Intestinal transit:** Undigested residue (fiber) and the physical volume of food remain in the digestive tract.

Your weight can increase by 1.5 kg from one day to the next without you having gained a single gram of fat. For this reason, clinical decisions within the calculation tool are made by evaluating weekly and biweekly trends, not 24-hour fluctuations. Anxiety arises from a lack of data; by recording precise metrics (weight, TDEE, intake, and circumferences), anxiety is replaced by methodical control. Ultimately, success in body recomposition is the triumph of mathematics over emotion.

CONCLUSION

MASTERY OF YOUR PHYSIOLOGY

The health and wellness industry has historically thrived by keeping people in a state of perpetual confusion, selling temporary solutions based on emotional restriction, fear of certain macronutrients, and biological ignorance. The Re/Built Health system was designed to break that cycle through the irrefutable application of data science and clinical thermodynamics.

Throughout this manual, we have dismantled nutritional dogmas to reveal how the body truly operates. The loss of adipose tissue and preservation of muscle mass are not random events, nor do they depend on esoteric factors or an inexhaustible supply of willpower. They are the strictly mathematical result of consistently auditing energy balance and ensuring an anabolic environment through precise protein intake.

The calculation tool that accompanies this methodology is not merely a dietary journal; it is the command center of body recomposition. By automating biometric equations, isolating Lean Body Mass (LBM), and processing real-world effort telemetry through technology, the system eliminates operational friction and decision fatigue. It transforms the anxiety of intuitive eating into predictable, calculated execution. The user provides the discipline of tracking; the algorithm returns absolute control over physical structure.

Success in this process does not require extreme motivation or punitive diets. It requires methodological consistency, precise data collection, and the clinical patience necessary to allow cellular biology to respond to a sustained stimulus over time. Control over health and body composition ceases to be an estimate based on hope and becomes a mathematical certainty.

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